CropVault

Decentralized Parametric Crop Insurance Protocol on Botchain

Whitepaper v1.0 | Botchain Agriculture Series | November 2026

1. Executive Summary

CropVault is a decentralized parametric crop insurance protocol on Botchain. Farmers purchase coverage against weather and pest events by paying BOT premiums. Payouts are triggered by on-chain oracle events and approved by a decentralized assessor network. A community staking vault backs all coverage, with stakers earning 70% of premium income as yield.

Only 3% of smallholder farmers in Africa have access to formal agricultural insurance. CropVault removes the need for traditional underwriters, claims adjusters, and paper documentation by replacing them with transparent smart contracts and cryptographic evidence (IPFS).

2. Problem Statement

2.1 Climate Risk is Uninsured

Droughts, floods, and pest outbreaks destroy billions of dollars of smallholder crops annually. Without insurance, a single weather event can drive a farming family into poverty. Traditional insurers avoid rural smallholders due to high administrative costs and low premiums.

2.2 Claims Fraud and Opacity

In the rare cases where insurance exists, lengthy claims processes, opaque assessments, and corruption mean payouts take months or never arrive. Farmers distrust the system.

3. Protocol Design

3.1 Coverage Types & Premiums

Coverage Type	Annual Premium Rate	Coverage Ratio	Use Case
Drought	3.0%	80% of crop value	Dry season crop failure
Flood	2.5%	80% of crop value	Riverine and flash flooding
Frost	2.0%	80% of crop value	Highland temperature drops
Pest	1.5%	80% of crop value	Locust, armyworm outbreaks
Comprehensive	6.0%	80% of crop value	All-risk combined coverage

3.2 Policy Lifecycle

- Farmer selects coverage type, declares crop value, and sets duration (30-365 days)
- Smart contract calculates premium: (cropValue x rate x days) / 365
- Farmer pays premium in BOT; policy minted on-chain with coverage = 80% of crop value
- 70% of premium distributed to stakers; 30% held in reserve pool
- If a weather event occurs, farmer files a claim with IPFS evidence hash
- Assessor (human or oracle) reviews and approves or rejects the claim
- Approved claims are paid automatically from the vault balance

3.3 Community Staking Vault

Anyone can stake BOT into the CropVault to back insurance policies. Stakers earn yield from premiums and bear loss risk if claims exceed the pool. A 7-day unstake lock prevents bank-run dynamics. The accRewardPerStake accumulator ensures precise, gas-efficient yield distribution.

3.4 Oracle and Assessor Architecture

In v1, a trusted assessor multisig (3-of-5) reviews claims within 48 hours. Evidence is submitted as IPFS CIDs (photos, sensor data, satellite imagery). In v2, Botchain oracles will auto-approve claims when oracle-reported event severity exceeds a configurable threshold for the farm location, enabling fully parametric payouts.

4. Milestones & Activities

Milestone	Activity	Timeline	KPI
M1 – Contract Deploy	Deploy CropVault.sol on Botchain testnet; full test suite	Month 1	Audited contract
M2 – Vault Bootstrap	Recruit 10 stakers; seed vault with 50 BOT initial liquidity	Month 2	50 BOT TVL
M3 – Pilot Policies	Issue 20 pilot policies to partner farmers in Oyo and Kano	Months 2-3	20 active policies
M4 – First Claim Cycle	Process first batch of claims; measure approval time	Months 3-4	<48hr claim resolution
M5 – dApp Launch	Launch CropVault frontend; policy purchase and claims UI	Month 3	Live dApp on Botchain

M6 – Assessor Network	Certify 5 regional assessors; deploy assessor portal	Month 4	5 active assessors
M7 – Oracle Integration	Integrate Botchain weather oracle for parametric triggers	Months 5–6	First oracle-triggered payout
M8 – Mainnet Scale	Launch to 500 policies; 230 BOT vault target	Month 6	500 policies, 230 BOT TVL

5. Solvency & Risk Controls

- Solvency check: vault must cover (totalStaked + premiumPool >= totalCoverage) before issuing new policy
- 7-day unstake lock prevents sudden liquidity drain
- 80% payout cap means the vault retains 20% of each loss event
- Reserve pool (30% of premiums) acts as first-loss buffer before staker capital
- Assessor multisig prevents single-point-of-failure fraud

6. Revenue Model

The protocol does not charge a management fee. 70% of premiums go to stakers, 30% is retained as a reserve. Assessors are compensated from the reserve fund at a flat rate per claim reviewed. Governance may adjust these ratios via DAO vote in future versions.

7. Conclusion

CropVault democratises access to agricultural insurance by replacing costly underwriters and opaque claims processes with transparent Botchain smart contracts. Every premium, coverage amount, claim decision, and payout is publicly verifiable. For smallholder farmers facing increasing climate volatility, CropVault represents the most accessible safety net ever built — available to anyone with a Botchain wallet.